IT599 Unit 2 Project Implementation Plan

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Remote Initiative for a VoIP Infrastructure

Introduction

The following implementation plan was created to provide a more in-depth look into the remote imitative for VoIP Fan executive officers and other sponsors within the project. This initiative hopes to improve staff moral as well as the reputation of the organization by taking the necessary actions to allow employees to work from remote locations of their choosing. The following sections will outline the budget for this project, stakeholders, work breakdown structure, estimated expenses, risk management plan, quality plan, and communications plan.

Overview of the Project

VoIP Fan will be allowing 200 staff members from the support department to work remotely. Of the 200 employees, it is estimated that 60 will require replacement routers and extended ethernet cables that will be provided by the organization. A network engineer will be responsible for helping each of the 200 employees with assessing their network to ensure they experience minimal issues with connection quality. This will involve assessing each network's bandwidth, connectivity, and equipment configuration. Web cams will be installed for all remote staff prior to them working remotely to ensure they have a means of attending any team or personal work meetings. To allow staff to access the necessary resources in a secure manner all computers will also have software installed for access to the organization's virtual private network and a mobile device management tool. The MDM will help VoIP Fan prevent inappropriate downloads or actions from being performed so that the organization, employees, and customer data remain secure.

Stakeholders

Stakeholders	Responsibilities
Project manager **Ms. Ainslie**	• Develop and manage project budget. • Develop and manage project schedule. • Develop a communications plan. • Document any changes to project, budget, schedule, or staff members. • Hold weekly status meetings to discuss and document successes or failures. • Maintain control of the project to avoid scope creep. • Assess and document project results at conclusion.
Systems Analyst **Mr. Lyst**	• Analyze the existing network topology and security measures. • Research potential solutions. • Evaluate infrastructure needs for accessibility and security. • Develop documentation for new employee setup. • Assist with training remote employees.
Software Engineer **Mr. Micro**	• Confirm new software products are compatible with existing software. • Install new software on remote employee devices. • Confirm new software is functional.
Network Engineer **Mrs. Wan**	• Ensure the organization has the necessary hardware to allow staff to work remotely. • Analyze the staff home networks to ensure they are compatible with VoIP. • Document any required changes to home network configurations or hardware. • Assist with making minor configuration changes with remote staff approval.
Hardware Technician **Mr. Fixit**	• Assess computer components for failures. • Update computer operating system and drivers. • Install webcams and confirm they are properly configured.
Quality Assurance **Ms. Perfect**	• Develop a plan that outlines what will be tested and how it will be tested. • Have users who are unfamiliar with the system test it to determine what changes or training may be needed. • Perform penetration testing on the remote user's devices to ensure the security measures remain operational in the office and at the home network. • Analyze the reports of all tests. • Report any negative discoveries promptly. • Develop an action plan in the event of failure.

Table 1. Stakeholder positions and responsibilities for the remote staff initiative.

Work Breakdown Structure

ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names	Milestone	Cost
1		**Analysis**	**4 days**	**Mon 2/22/2**	**Thu 2/25/21**		**Mr. Lyst**	**Yes**	**$2,240.00**
2		Analyze Hardware	2 days	Mon 2/22/2	Tue 2/23/21		Mr. Lyst	No	$560.00
3		Analyze Software	2 days	Wed 2/24/2	Thu 2/25/21	2	Mr. Lyst	No	$560.00
4		**AUP**	**5 days**	**Fri 2/26/21**	**Thu 3/4/21**	**3**	**Ms. Ainslie**	**Yes**	**$1,600.00**
5		Create Contracts	2 days	Fri 2/26/21	Mon 3/1/21	1	Ms. Ainslie	No	$320.00
6		Contracts Signed	3 days	Tue 3/2/21	Thu 3/4/21	5	Ms. Ainslie	No	$480.00
7		**Prepare Hardware**	**9 days**	**Fri 3/5/21**	**Wed 3/17/2**	**6**	**Web Cams[60],Mr. Fi**	**Yes**	**$6,504.00**
8		Update Computers	5 days	Fri 3/5/21	Thu 3/11/21	6	Mr. Fixit	No	$640.00
9		Install web cams	4 days	Fri 3/12/21	Wed 3/17/2	8	Mr. Fixit	No	$512.00
10		**Prepare Software**	**10 days**	**Fri 3/5/21**	**Thu 3/18/21**	**6**	**VPN[200],MDM[200],**	**Yes**	**$24,360.00**
11		Install MDM	5 days	Fri 3/5/21	Thu 3/11/21	6	Mr. Micro	No	$1,680.00
12		Install VPN	5 days	Fri 3/12/21	Thu 3/18/21	11	Mr. Micro	No	$1,680.00
13		**Prepare Network**	**26 days**	**Fri 3/5/21**	**Fri 4/9/21**	**6**	**Cat5[60],Routers[60],**	**Yes**	**$27,886.00**
14		**Network Check**	**5 days**	**Fri 3/5/21**	**Thu 3/11/21**	**6**	**Mrs. Wan**	**No**	**$1,640.00**
15		Bandwidth	5 days	Fri 3/5/21	Thu 3/11/21			No	$0.00
16		Latency	5 days	Fri 3/5/21	Thu 3/11/21			No	$0.00
17		Packet loss	5 days	Fri 3/5/21	Thu 3/11/21			No	$0.00
18		**Configuration**	**7 days**	**Fri 3/12/21**	**Mon 3/22/2**	**14**	**Mrs. Wan**	**No**	**$2,296.00**
19		Disable SIP ALG	5 days	Fri 3/12/21	Thu 3/18/21			No	$0.00
20		Enable Multicast	7 days	Fri 3/12/21	Mon 3/22/2			No	$0.00
21		Confirm No Double Nat	5 days	Fri 3/12/21	Thu 3/18/21			No	$0.00
22		Firewall Rules	5 days	Fri 3/12/21	Thu 3/18/21			No	$0.00
23		Install New Routers	14 days	Tue 3/23/21	Fri 4/9/21	18	Mrs. Wan	No	$4,592.00

Figure 1. Part one of the work breakdown structure for the remote staff initiative.

ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names	Milestone	Cost
17		Packet loss	5 days	Fri 3/5/21	Thu 3/11/21			No	$0.00
18		**Configuration**	**7 days**	**Fri 3/12/21**	**Mon 3/22/2**	**14**	**Mrs. Wan**	**No**	**$2,296.00**
19		Disable SIP ALG	5 days	Fri 3/12/21	Thu 3/18/21			No	$0.00
20		Enable Multicast	7 days	Fri 3/12/21	Mon 3/22/2			No	$0.00
21		Confirm No Double Nat	5 days	Fri 3/12/21	Thu 3/18/21			No	$0.00
22		Firewall Rules	5 days	Fri 3/12/21	Thu 3/18/21			No	$0.00
23		Install New Routers	14 days	Tue 3/23/21	Fri 4/9/21	18	Mrs. Wan	No	$4,592.00
24		**Testing**	**10 days?**	**Fri 3/12/21**	**Thu 3/25/21**		**Ms. Perfect**	**Yes**	**$5,760.00**
25		**Test MDM**	**3 days**	**Fri 3/12/21**	**Tue 3/16/21**	**11**	**Ms. Perfect**	**No**	**$864.00**
26		Test Alerts	3 days	Fri 3/12/21	Tue 3/16/21			No	$0.00
27		Test compromising MDM	3 days	Fri 3/12/21	Tue 3/16/21			No	$0.00
28		**Test VPN**	**1 day?**	**Wed 3/17/2**	**Wed 3/17/2**	**25**	**Ms. Perfect**	**No**	**$288.00**
29		Resource Accessibility	1 day?	Wed 3/17/2	Wed 3/17/2			No	$0.00
30		Connectivity	1 day?	Wed 3/17/2	Wed 3/17/2			No	$0.00
31		Test Webcams	3 days	Thu 3/18/21	Mon 3/22/2	9,25	Ms. Perfect	No	$864.00
32		Test Connectivity	3 days	Tue 3/23/21	Thu 3/25/21	14,31	Ms. Perfect	No	$864.00
33		Test Access Controls	0 days	Thu 3/25/21	Thu 3/25/21	32	Ms. Perfect	Yes	$0.00
34		**Assessments**	**5 days**	**Fri 3/26/21**	**Thu 4/1/21**	**24**	**Ms. Ainslie**	**Yes**	**$1,600.00**
35		QA Review	2 days	Fri 3/26/21	Mon 3/29/2	24	Ms. Ainslie	No	$320.00
36		Stakeholder Notice	1 day	Tue 3/30/21	Tue 3/30/21	35	Ms. Ainslie	No	$160.00
37		Final Status	1 day	Wed 3/31/2	Wed 3/31/2	36	Ms. Ainslie	No	$160.00
38		Lessons Learned	1 day	Thu 4/1/21	Thu 4/1/21	37	Ms. Ainslie	No	$160.00

Figure 2. Part two of the work breakdown structure for the remote staff initiative.

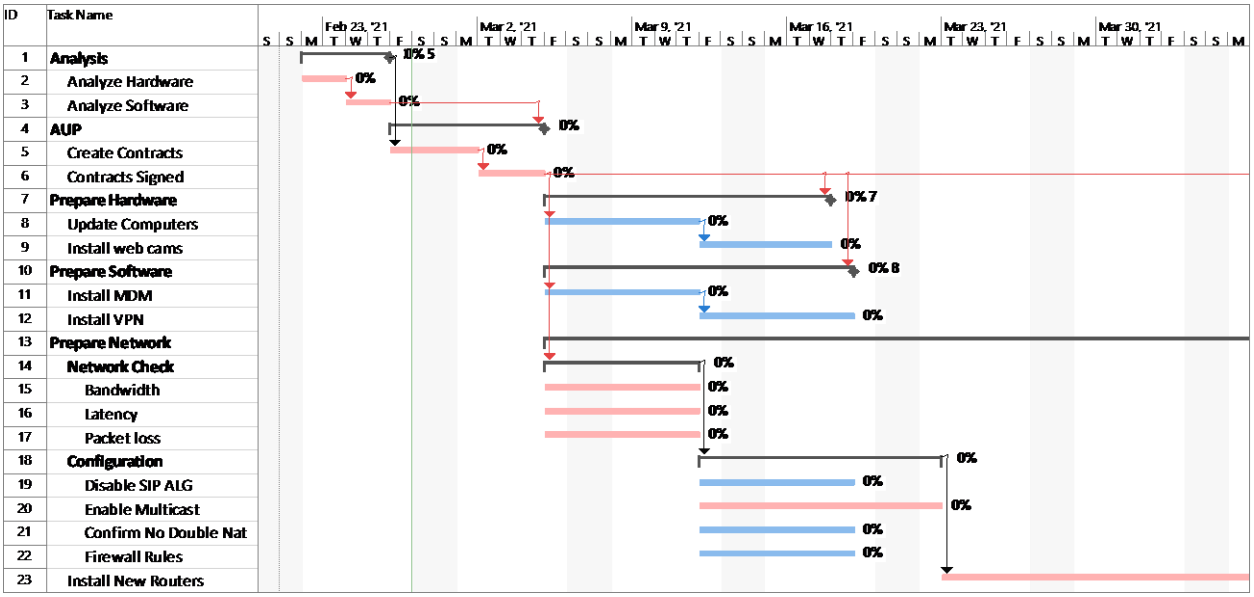


Figure 3. Part one of the work breakdown structure schedule and dependencies for the remote staff initiative.

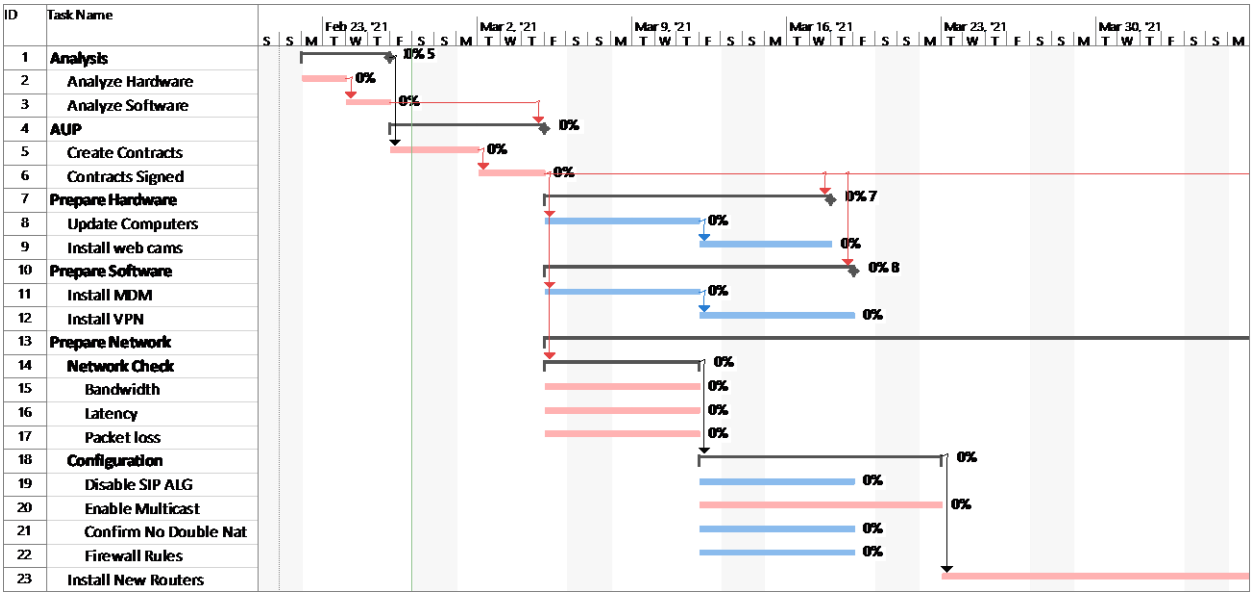


Figure 4. Part two of the work breakdown structure schedule and dependencies for the remote staff initiative.

Cost Details

The projected cumulative cost for this project is $71,934. This total is broken down in the following charts by weekly projections, labor , material, and milestone expenses.



Figure 5. Weekly and cumulative costs for the remote staff initiative.

Name	Cost	Work	Standard Rate
Mr. Lyst	$2,240.00	64 hrs	$35.00/hr
Mrs. Wan	$15,744.00	384 hrs	$41.00/hr
Mr. Micro	$6,720.00	160 hrs	$42.00/hr
Ms. Perfect	$8,928.00	248 hrs	$36.00/hr
Ms. Ainslie	$3,200.00	160 hrs	$20.00/hr
Mr. Fixit	$2,432.00	152 hrs	$16.00/hr

Table 3. Labor hours, rates, and total costs for the remote staff initiative.

Name	Cost	Units	Standard Rate
Web Cams	$14,000.00	200	$70.00
VPN	$8,040.00	200	$40.00
MDM	$9,600.00	200	$48.00
Routers	$8,940.00	60	$149.00
Cat5	$1,890.00	60	$31.50

Table 3. Material units, rates, and total costs for the remote staff initiative.

Milestone Name	Cost	Task ID
Analysis	$1,120.00	1
AUP	$800.00	4
Prepare Hardware	$5,480.00	7
Prepare Software	$21,000.00	10
Prepare Network	$18,702.00	13
Testing	$4,032.00	24
Assessments	$800.00	34

Table 4. Milestone expenditure totals based on milestones for the remote staff initiative.

Risk Management Plan

Risk	Impact	Mitigation Strategy
Internet Outages	Staff is unable to assist customer(s).	Hotspots for remote staff.
ISP doesn't support company provided routers	Pre-purchased routers would need to be replaced.	Contact ISP prior to purchasing routers.
Accessibility Issues	Staff is unable to assist customer(s).	Prepare a list of end-user access requirements prior to setting access controls. Confirm these resources are all accessible during VPN testing.
Network configuration delays	Staff cannot work remotely until network is properly configured. Delays also will increase labor costs.	Establish a schedule with employees who wish to work remotely and add the network check requirement in the mandatory AUP. Define a checklist to expedite network checks and provide this to staff prior to the scheduled session.

Table 5. A risk assessment and mitigation plan for the remote staff initiative.

Quality Plan

Quality reviews will be completed throughout the project to mitigate potential delays and avoid unnecessary re-work by team members.

Deliverable	Acceptance Criteria
Hardware Preparation	All computers must have up to date operating systems and drivers with web cams installed and functional.
Software Preparation	All computers must have MDM and VPN software installed to allow secure access to company resources. This must be installed in such a way that employees cannot disable the software or compromise company resources.
Network Preparation	Home networks must have at least 10 MB up and down speeds, latency not exceeding 150ms, and no packet loss. All routers must have SIP ALG disabled, Multicast enabled, firewall rules configured, and no more than one device performing NAT. Any routers that cannot be configured to meet these needs must be replaced with equipment that can.

Table 6. This table defines the deliverables for the remote staff initiative along with the related acceptance criteria.

Communication Plan

Communication	Participants	Method	Timing	Prepared by
Kick-Off Meeting	All Stakeholders	In person meeting.	One-time event	Project Manager
Status Reports	All Stakeholders	E-mail	Weekly	Project Manager
Milestone Review	All Stakeholders	Virtual Meeting	Milestone completion	Project Manager
QA Review	One-on-one each stakeholder	Phone call		Project Manager
Lessons Learned	All Stakeholders	In person meeting	One-time event	Project Manager

Table 7. This table illustrates the expected meetings, participants, meeting method, and meeting schedules.

Conclusion

The purpose of this paper is to outline the anticipated expenses, stakeholders, tasks, and risks for the remote staff initiative. Expenses were broken down to stress the project costs of labor, materials, and the total costs incurred for each milestone. Details included throughout the implementation include project milestones, deliverables, and acceptance criteria. The implementation plan was concluded with a communication plan to outline how and what meetings will occur to exchange information about the project.